

Numerical Simulation of Accretion onto Compact Objects with Hard Surface

Aswin R Nair

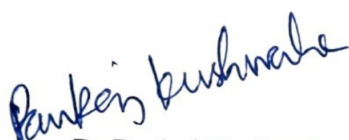
*A dissertation submitted for the partial fulfilment of BS-MS dual degree in
Science*



**Indian Institute of Science Education and Research, Mohali
April 2025**

Certificate of Examination

This is to certify that the dissertation titled **Numerical Simulation of Accretion onto Compact Objects with Hard Surface** submitted by **Aswin R Nair** (Reg. No. MS20028) for the partial fulfillment of BS- MS Dual Degree programme of the institute, has been examined by the thesis committee duly appointed by the institute. The committee finds the work done by the candidate satisfactory and recommends that the report be accepted.


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Declaration

The work presented in this dissertation has been carried out by me under the guidance of Dr.Indranil Chattopadhyay from Aryabhata Research Institute of Observational Sciences (ARIES), Nainital and Dr.Pankaj Kushwaha at the Indian Institute of Science Education and Research, Mohali.

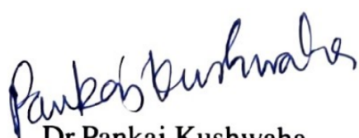
This work has not been submitted in part or in full for a degree, a diploma, or a fellowship to any other university or institute. Whenever contributions of others are involved, every effort is made to indicate this clearly, with due acknowledgement of collaborative research and discussions. This thesis is a bonafide record of original work done by me and all sources listed within have been detailed in the bibliography.



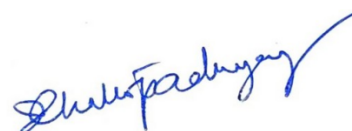
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In my capacity as the supervisor of the candidates project work, I certify that the above statements by the candidate are true to the best of my knowledge.



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Abstract

Accretion onto a weakly magnetised compact object is studied analytically as well as numerically. We use pseudo-Newtonian potential to mimic strong gravity. Relativistically correct Chattopadhyay-Ryu (CR) equation of state is used to describe the thermodynamics of the relativistic gas. We analytically derive the conditions which demarcate accretion onto a compact object with and without a hard surface. We then regenerate these solutions numerically and at the same time ensuring the formation of a steady terminal shock close to the surface of the star. It is expected that if the flow is transonic then at least one terminal shock will form, but can additional shocks form? We investigate the scenario where different kinds of these additional shocks form.

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Chapter 1

Introduction

1.1 Compact Object Accretion

Compact objects are a class of celestial bodies that have extreme densities. These objects usually form at the end of stellar lifecycles, usually from the result of a spectacular explosion or grand fiery death. The main candidates in this class are - Black holes, neutron stars and white dwarfs. When a star reaches the end of its stellar evolution, it either sheds off its outer layers, leaving behind a degenerate core or undergoes a supernova explosion. Lighter stars end up as white dwarfs, where their cores are supported by electron degeneracy pressure; heavier stars end up as neutron stars with cores supported by neutron degeneracy pressure, and even heavier stars collapse completely into a black hole. ([Shapiro and Teukolsky 1983](#); [Camenzind 2007](#))

Compact objects like neutron stars of masses $1\text{-}2\ M_{\odot}$ are compacted into a radius of ~ 10 km; for comparison, the radius of the sun is at $\sim 10^6$ km.

Table 1.1: A comparison of the compactness ratio of different celestial objects

Object	Mass (M_*)	Radius (R_*)	$\mathcal{C}_{\mathcal{R}} = R_s/R_*$
Black hole (maximally rotating)	$\gtrsim 3 - 10^9 M_{\odot}$	$r_s \simeq 3(M_*/M_{\odot})$ km	1
Neutron star	$\sim 1 - 3 M_{\odot}$	~ 10 km	$0.16 - 0.3$
White dwarf	$\sim 1 M_{\odot}$	$\sim 7 \times 10^3$ km	2×10^{-4}
Sun-type star	$1 M_{\odot}$	$\sim 7 \times 10^5$ km	4.2×10^{-6}

Table 1.1 presented in [Sarkar 2022](#) shows a comparison of the "compactness ratio" defined as the ratio of the Schwarzschild radius of the object to the radius of the surface; it is essentially a comparison of their densities. The table compares the three compact objects with a sun-like star; It provides a sense of how compact these bodies are.

Such intense densities grant these objects very steep gravitational potentials, which makes their accretion tendencies very prominent. As matter is accreted from far-off distances and then pulled onto these objects, their intense gravity squeezes them into ever-increasing densities while it falls at ever-increasing velocities. Most of the gravitational potential which the flow possesses at this point is often emitted as very energetic electromagnetic radiation like X-rays and Gamma rays([Frank et al. 2002](#)). Because of their steep gravitational potential, accretion flows for these objects may reach supersonic velocities near the stellar object. This phenomenon of transonic flows, where flows transition from subsonic to supersonic (or vice-versa in case of outflows), has been extensively studied in the literature and is an integral part of compact object accretion dynamics ([Bondi 1952](#); [Shapiro and Salpeter 1975](#); [Liang and Thompson 1980](#); [Abramowicz and Zurek 1981](#); [Chakrabarti 1989](#)).

In rotating flows, the accreted matter forms accretion disks. Various models have been proposed for such disks with the major proponents being the α -disk by [Shakura and Sunyaev 1973](#), Novikov-Thorne model ([Novikov and Thorne 1973](#)), the thick disk model by [Fishbone and Moncrief 1976](#); [Abramowicz et al. 1980](#); [Paczynski and Wiita 1980](#), etc. However, most of these disk models assume viscosity to be significant in the flow; the viscous forces act as a medium of angular momentum transport in these models.

At the innermost part of the accretion flow, matter starts to interact with the compact object itself. For black holes, the flow plunges into the event horizon, never to be seen again. However, compact objects like neutron stars and white dwarfs have a hard surface beyond which no matter can go. This presents complexities in the accretion flow that are not seen in black hole accretion([Shapiro and Salpeter 1975](#)). In addition to the hard surface-flow interaction problems, the intense magnetic fields of these compact objects also modify the dynamics of the accretion flow and present unique properties to the observer. Perhaps one of the most interesting instances of this is the very precise (in timing) radio pulses seen from fast-rotating neutron stars called Pulsars. In this thesis, we will focus on the accretion of neutron stars, even

though white dwarfs are also compact objects with hard surfaces. We are interested in probing the specific properties of neutron star accretion where the magnetic fields are comparatively weaker.

1.2 Weakly magnetised neutron stars and their binary systems

Neutron stars harbour one of the strongest magnetic fields in the universe, with field strengths reaching upwards of 10^{12} Gauss. By comparison, the magnetic field of the sun is of the order of ~ 1 Gauss, and that of the Earth is $\sim 10^{-5}$ Gauss. However, not all neutron stars are equal in terms of their powerful magnetic fields; neutron stars can be roughly classified into two groups depending on how their magnetic fields compare to their gravitational potential - Strongly magnetised neutron stars and Weakly magnetised neutron stars ([Lewin and van der Klis 2006](#)).

Strongly magnetised neutron stars have stellar fields in the range $\sim 10^{12} - 10^{14}$ Gauss. These neutron stars have magnetic fields that are strong enough to disrupt the accretion flow and force the gas to follow the magnetic field lines ([Ghosh and Lamb 1979](#)). These rapidly rotating neutron stars also produce strong radio pulses that are seen as pulsar sources from Earth and are often seen to be part of high-mass X-ray Binaries (HMXBs). Weakly magnetised neutron stars have stellar fields in the range $\sim 10^8 - 10^{10}$ Gauss. Their magnetic fields are not strong enough to disrupt the accretion flow, and thus, matter can reach all the way to the surface of these stellar objects ([Psaltis and Chakrabarty 1999](#)). These neutron stars are often part of Low-Mass X-ray Binaries (LMXBs) and primarily emit in X-rays. The salient feature in the X-ray emissions of these systems is their variability that is often seen as thermonuclear bursts ([Woosley and Taam 1976](#)) or the less understood Quasi-Periodic Oscillations in the Khz range ([van der Klis 2000](#)).

In weakly magnetised systems, the absence of magnetic funnelling means that the accretion flow interacts directly with the stellar surface, producing a boundary layer—a narrow region where the accretion disk abruptly decelerates to match the star’s rotational velocity and reaches rest on the surface ([Inogamov and Sunyaev 1999](#); [Inogamov and Sunyaev 2010](#)). This collision converts kinetic energy into heat, generating temperatures exceeding 10^8 K and emitting X-rays that dominate

the system's radiative output ([Popham and Sunyaev 2001](#); [Grebenev and Sunyaev 2002](#)).

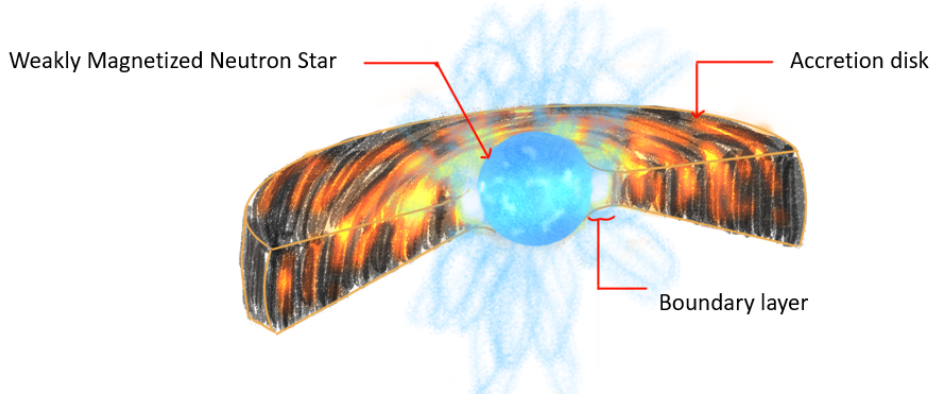


Figure 1.1: An illustration showing a weakly magnetised neutron star with its accretion disk cut away to reveal the boundary layer present where the accretion flow meets the surface.

1.3 Problem motivation

As mentioned in section 1.1, accretion flows accelerate to supersonic velocities near the compact object. For Black holes, this is not an issue since the flow can pass through the event horizon unhindered; however, for compact objects with hard surfaces like neutron stars and white dwarfs, the flow is stopped at the surface. This creates a problem where interactions between the supersonic flow and the surface are unclear; the accreting flow needs to somehow brake from supersonic speeds down to zero at the surface. The main mechanism which allows such flow breaking is through shocks in the accretion flow and further slowing by dissipation in the Boundary layer in the context of weakly magnetised neutron stars.

This thesis aims to study such accretion onto weakly magnetised neutron stars. Studies have been done on neutron star accretion in various capacities, like the work of [Shapiro and Salpeter 1975](#), which focuses on the adiabatic shock fronts that form near the surface, [Mukhopadhyay 2002](#), which considers cooled flow solutions with both centrifugal barrier shocks and surface-induced shocks. Apart from these analytical works numerical simulation of such flows also exist like the work of [Bhattacharjee and Chakrabarti \(2019\)](#) which considers 2D SPH simulation of neutron star accretion flows showing centrifugal and surface shocks.

The work that will be done in this thesis will feature a semi-analytical analysis of

accretion onto weakly magnetised neutron stars in the hydrodynamic regime with fully relativistically correct EoS presented in [Chattopadhyay and Ryu 2009](#) that has a variable adiabatic index. We will explore the possibility of there being multiple shocks in the flow apart from the standard centrifugal and near-surface shocks that are usually presented. We then finally verify which of these shocks are viable using a 1-D Eulerian hydrodynamical simulation. We try to model the boundary layer seen in such neutron Stars and use its properties to try and stabilise shocks in the presence of a fully reflecting inner boundary at the surface of the neutron star.

The thesis is organised as follows: In Chapter [2](#), the analytical study of the flow is presented and sections [2.1](#) - [2.5](#) outline the derivation of equations needed to carry out the analysis. Section [2.6](#) outlines the parameters that are used for the study, and in section [2.7](#), we present the results from the analytical study that was done. In Chapter [3](#), the numerical study of the flow is presented. Section [3.1](#) has the details of the simulation, and sections [3.2](#) and [3.3](#) present the results of the simulation for black hole and neutron star simulations, respectively. Chapter [4](#) ends with conclusions and discussions about the study.

Chapter 2

Analyzing accretion flows

The analytical considerations that go into simulating accretion flow onto a compact object with a hard surface are outlined in this chapter. The analytical solutions that are obtained here are used as initial conditions for the simulation. However, there is no information about a hard surface that is included in the following solutions, and for all intents and purposes, we are analysing black hole solutions. Nonetheless, it is useful to consider these solutions as the starting point and use hydrodynamical simulations to see how having a hard surface will affect the flow and cause it to deviate from typical black hole accretion solutions.

2.1 Geometry of the system

The compact object is placed at the origin, around which an accretion disk is present. The accretion is assumed to be conically shaped, which makes it axisymmetric around the Z axis, with the disk lying parallel to the XY plane. All calculations are carried out in spherical coordinates that involve R, θ and ϕ .

The analysis in this thesis is done in the radial direction only (1-dimensional). For ease of calculation, henceforth, all units will be converted as follows:

$$x \equiv r/r_s \text{ (distance)}$$

$$v \equiv u/c \text{ (velocity)}$$

$$a \equiv c_s/c \text{ (speed of sound)}$$

$$\lambda \equiv l/r_s c \text{ (angular momentum)}$$

where r_s is the Schwarzschild radius: $r_s = 2GM/c^2$ and c is the speed of light. Further we set $2GM = c = 1$ and $M \equiv M_{bh}$ where M_{bh} is the mass of the black hole.

2.2 Assumptions in the analysis

Since it would be very complex to consider every detail involved in modelling accretion onto a stellar surface, we work under some assumptions to simplify the system that we can analyse.

The first of the key assumptions we make is that the accretion flow is purely hydrodynamical, and no magnetic field is considered in the analysis. When analysing typical neutron star accretion flows, magnetic fields are a prominent factor. However, we can safely disregard magnetic fields if they are weakly magnetised neutron stars, as the magnetic pressure would not be strong enough to oppose gas pressure and thus would not affect the gas dynamically.

Let us justify this argument using an "order of magnitude" analysis. For our argument to be valid, the magnetic energy density needs to be less than the total energy density of the fluid.

$$\frac{B^2}{8\pi} \lesssim \frac{1}{2}\rho v^2 + \rho e + \rho\Theta$$

where e is the specific internal energy and $\rho\Theta$ represents gas pressure, the definition of these terms are elaborated in coming sections. We also define density $\rho = \frac{\dot{M}}{vr^2}$, where r could be considered the Alfvén radius and $\dot{M}$ is the accretion rate. Plugging in the typical values (in cgs units) seen in the accretion flows at a rate $\dot{M} \sim 10^{18}g/s$, which is near the Eddington limit and setting the Alfvén radius to stellar radius $r \sim 3r_s$ we get;

$$\frac{B^2}{8\pi} \lesssim \frac{10^{15} \times 10^{-3}}{10^{-1} \times (3 \times 4 \times 10^3)^2} [0.5 \times (10^{-1})^2 + 1 + 10^{-1}] \times 3 \times 10^8$$

$$B \lesssim 10^8 G$$

which is the typical magnetic field strength for weakly magnetised Neutron Stars ([Salvo et al. 2024](#)). This assumption also greatly simplifies the analysis and simulation since magnetohydrodynamic systems are much more complex and computationally intensive.

Another key assumption is that the accretion flow is non-viscous. This marks a stark difference from viscous flow models for accretion by compact objects like the α -disk model presented by [Shakura and Sunyaev 1973](#). The α -disk model relies on viscosity generated through hydrodynamic and/or magnetic turbulence in the

flow. However, hydrodynamic turbulence might not be efficient for trans-sonic flows (Lesur and Longaretti 2005; Martin et al. 2019).

In this thesis, it is instead assumed that, due to negligible viscosity in trans-sonic flows (magnetic instabilities are not considered), no viscous forces act on the fluid. The absence of viscosity in the flow leads to the absence of any meaningful mechanisms that contribute to angular momentum transport within the accretion flows (which is usually mediated by viscous torque). Thus, angular momentum is assumed to be constant. The flow is also assumed to be adiabatic, thus no energy escapes the system.

2.3 Hydrodynamics equations

The fundamental equation used in our hydrodynamical analysis is the Euler equation evaluated in spherical coordinates;

$$\rho \left[\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \vec{\nabla}) \vec{u} \right] = -\vec{\nabla} P - \rho \vec{f} \quad (2.1)$$

and the continuity equation;

$$\frac{\partial(\rho)}{\partial t} + \vec{\nabla} \cdot (\rho \vec{u}) = 0 \quad (2.2)$$

where the symbols have their usual meaning.

Since we are working with a one-dimensional analysis, we set the flow to have a velocity $\vec{u} = \{v_r, 0, v_\phi\}$. This constrains the flow to the XY plane; since the angular momentum is constant, it also makes the flow axisymmetric. Using the preceding simplification and restricting this analysis to steady-state solutions, we can transform the above vector equation into a simpler radial form:

$$v \frac{dv}{dx} + \frac{1}{\rho} \frac{dp}{dx} - \frac{\lambda^2}{x^3} + \frac{1}{2(x-1)^2} = 0 \quad (2.3)$$

where the symbols represent converted units as defined in Section 2.1.

Another important equation used in our hydrodynamical analysis is the First Law of Thermodynamics, which represents the conservation of energy;

$$\frac{d\bar{e}}{dx} - \frac{p}{\rho^2} \frac{d\rho}{dx} = 0 \quad (2.4)$$

The last term in equation 2.3, representing the gravitational potential, is derived from the Paczyński-Wiita potential (Paczyński and Wiita 1980);

$$\Phi_{PW} = -\frac{GM}{r - r_s} \equiv -\frac{1}{2(x - 1)}$$

This pseudo-Newtonian potential allows us to work in a Newtonian framework while mimicking the effects of General Relativity. Since Neutron Stars have relatively extreme gravitational potentials, it is necessary to include the effects of General Relativity.

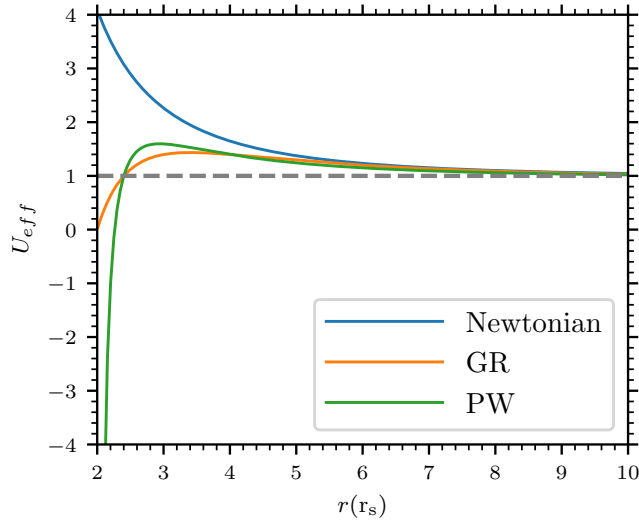


Figure 2.1: Comparison of different gravitational potentials in an effective potential $U_{(eff)}$ vs radial distance plot at an angular momentum of $\lambda = 1.68$. Namely, Newtonian potential, General Relativistic (GR) potential and Paczyński-Wiita (PW) potential are shown. The dashed line represents the rest of the mass-energy in non-dimensional units.

In Figure 2.1, we can see that the effective potential energy of a body keeps increasing to infinity as it gets closer to the centre of a Newtonian gravitational potential. The forms of the effective potentials plotted here are given below;

$$\begin{aligned} U_{\text{eff}}^{\text{Newt}} &= -\frac{1}{2x} + \frac{\lambda^2}{2x^2} + 1 \\ U_{\text{eff}}^{\text{PW}} &= -\frac{1}{2(x-1)} + \frac{\lambda^2}{2x^2} + 1 \\ U_{\text{eff}}^{\text{GR}} &= -\frac{2}{x} - \frac{2\lambda^2}{x^3} + \frac{\lambda^2}{x^2} + 1 \end{aligned}$$

where a factor of 1 has been added to the Newtonian and Paczyński-Wiita potentials to account for the rest mass energy term occurring in GR.

This caveat restricts any flow that does not lose its angular momentum, in hopes of reducing its energy enough to reach the centre. However, general relativistic or pseudo-Newtonian potentials like the Paczyński-Wiita potential allow the body to lose its energy as it goes near the centre while keeping a constant angular momentum and reaching the centre of the potential.

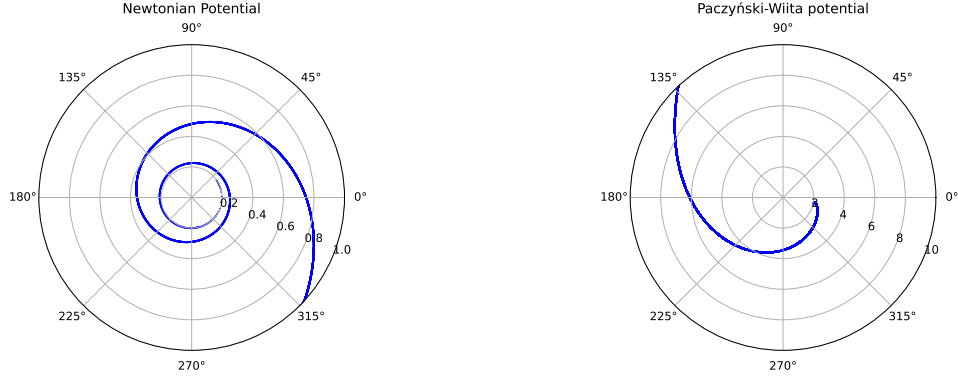


Figure 2.2: Streamlines showing the flow approaching $r = 0$ in different potentials. The flow winds up, approaching a circular stable orbit before the centre in the Newtonian potential (left) while the flow streams freely towards the centre in the Paczyński-Wiita potential (right)

One would notice that the scales are different in the two plots in Fig 2.2. This is because the Newtonian potential only allows flows with a minuscule amount of angular momentum ($\lambda = 0.16$ in the figure) to be in a bound orbit, which means the flow circulates at shorter distances. The Paczyński-Wiita potential allows flows to stream a much higher angular momentum ($\lambda = 1.7$ in the figure) but terminates at the event horizon at $r = r_s$ (due to coordinate singularity).

Equation (2.3) is the conservation of momentum in the differential form; likewise, equation (2.2) is the conservation of mass in the differential form. Converting these equations into the integral form gives us;

$$\mathcal{E} = \frac{v^2}{2} + h + \frac{\lambda^2}{2x^2} - \frac{1}{2(x-1)} \quad (2.5)$$

$$\dot{M}_{in} = 4\pi \cos \theta_0 \rho v x^2 \quad (2.6)$$

where h is the enthalpy (function of p and ρ) of the flow, $\mathcal{E}$ is called the Bernoulli parameter or the specific energy of the flow and $\dot{M}_{in}$ is the radial accretion rate, where θ_0 is the co-latitude angle.

To solve these equations, we need them in a closed system, which is achieved by defining a relation between p and other flow quantities. This is given by the Equation of State (abbreviated as EoS).

2.4 Equation of State: Variable Γ

The temperature of accretion flow can vary by multiple orders of magnitude from very large distances to near the stellar surface. This presents a problem for EoS that assumes a constant adiabatic index Γ since it will vary from thermally non-relativistic flow ($\Gamma \sim 5/3$) to thermally relativistic flow ($\Gamma \sim 4/3$). In the seminal paper by [Taub 1948](#), it is demonstrated that a relativistically correct EoS cannot work with this assumption. Thus, we use the Chattopadhyay & Ryu Equation of State (CR EoS) from [Chattopadhyay and Ryu 2009](#), which considers a multispecies ionised flow containing protons, electrons and positrons with a variable adiabatic index. The adiabatic index is dependent on the temperature, which can be determined through other relations and flow composition. [Vyas et al. 2015](#) shows that it is a valid approximation of earlier relativistically correct EoS, which was independently discovered by many others ([Chandrasekhar 1939](#); [Synge 1957](#); [Cox and Giuli 1968](#)).

This approximate EoS was developed to be easily implemented in a numerical calculation as earlier EoS were all modified Bessel's functions of different kinds, which slow down numerical calculations by a significant amount. A recently modified version of CR EoS was introduced in [Joshi et al. 2021](#), which is used in this analysis.

The CR EoS can be represented as;

$$e = \sum_i e_i = \sum \left[n_i m_i c^2 + p_i \left(\frac{9p_i + 3n_i m_i c^2}{3p_i + 2n_i m_i c^2} \right) \right]. \quad (2.7)$$

where the index i represents different species and c is used explicitly, elsewhere in our analysis c remains unity.

We define density as;

$$\rho = \sum n_i m_i = n_{e^-} m_{e^-} \left[2 - \xi \left(1 - \frac{1}{\eta} \right) \right] = n_{e^-} m_{e^-} \tau$$

where,

$$\begin{aligned}\tau &= \left[2 - \xi \left(1 - \frac{1}{\eta} \right) \right] \\ \xi &= \frac{n_{p^+}}{n_{e^-}} \\ \eta &= \frac{m_{e^-}}{m_{p^+}}\end{aligned}$$

and n_{p^+} , n_{e^-} , m_{e^-} and m_{p^+} are the proton number density, electron number density, electron rest mass and proton rest mass, respectively.

Thus, equation (2.7) can also be defined as;

$$e = \rho f \quad (2.8)$$

where,

$$f = 1 + (2 - \xi)\Theta \left[\frac{9\Theta + 6/\tau}{6\Theta + 8/\tau} \right] + \xi\Theta \left[\frac{9\Theta + 6/\eta\tau}{6\Theta + 8/\eta\tau} \right] \quad (2.9)$$

We then define a dimensionless temperature measure for the flow as follows;

$$\Theta = \frac{p}{\rho} \quad (2.10)$$

The specific enthalpy for this flow can be defined as;

$$h = \frac{e + p}{\rho} = f + \Theta \quad (2.11)$$

Consequently, the polytropic index can be given as;

$$N = \rho \frac{\partial h}{\partial p} - 1 = \frac{\partial f}{\partial \Theta} \quad (2.12)$$

$$N = 6(2 - \xi) \left[\frac{9\Theta^2 + 24\Theta/\tau + 8/\tau^2}{(6\Theta + 8/\tau)^2} \right] + 6\xi \left[\frac{9\Theta^2 + 24\Theta/(\eta\tau) + 8/(\eta\tau)^2}{\{6\Theta + 8/(\eta\tau)\}^2} \right] \quad (2.13)$$

And the adiabatic index is given as;

$$\Gamma = 1 + \frac{1}{N} \quad (2.14)$$

The speed of sound in this flow is defined as;

$$a^2 = \frac{\partial p}{\partial \rho} = \Gamma \Theta \quad (2.15)$$

Thus, we can describe a quantity called Mach number, which is defined as;

$$M = \frac{v}{a} \quad (2.16)$$

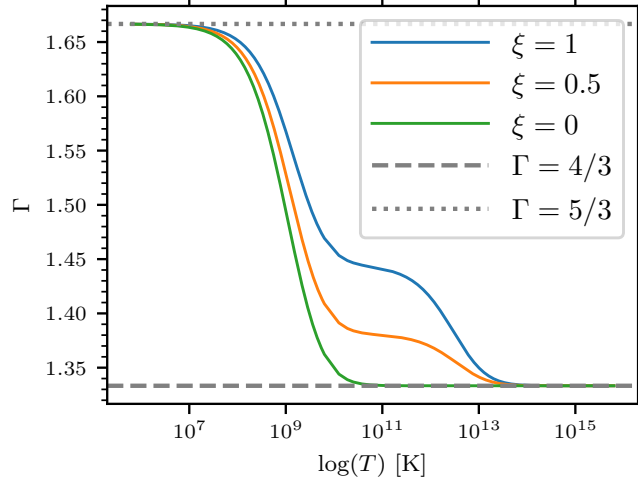


Figure 2.3: Adiabatic index varying with temperature at different composition proton-electron ratios (ξ). The adiabatic index saturates to 4/3 at high temperatures and saturates to 5/3 at low temperatures at different ξ values.

Integrating the conservation of energy given in equation (2.4) and using the definition of e (eq. (2.8)), f (eq. (2.9)) and relation between p and ρ (eq. (2.10)), we obtain;

$$\rho = \mathcal{K} \left(6\Theta + \frac{8}{\tau}\right)^{k_1} \left(6\Theta + \frac{8}{\eta\tau}\right)^{k_2} \Theta^{3/2} \exp(k_3) \quad (2.17)$$

where $\mathcal{K}$ is an integration constant and,

$$\begin{aligned} k_2 &= 3\xi/4 \\ k_1 &= 3(2 - \xi)/4 \\ k_3 &= -\frac{6}{\tau} \left[\frac{2 - \xi}{6\Theta + \frac{8}{\tau}} + \frac{\xi/\eta}{(6\Theta + \frac{8}{\eta\tau})} \right] \end{aligned}$$

The $\mathcal{K}$ obtained in equation (2.17) has information about the entropy of the flow; if the entropy is constant so is $\mathcal{K}$, thus we can define a quantity called the Entropic-accretion rate;

$$\dot{\mathcal{M}} = \frac{\dot{M}_{in}}{4\pi\mathcal{K} \cos \theta_0} = vx^2 \left(6\Theta + \frac{8}{\tau}\right)^{k_1} \left(6\Theta + \frac{8}{\eta\tau}\right)^{k_2} \Theta^{3/2} \exp(k_3) \quad (2.18)$$

2.5 Equations of motion

We derive the equations of motion in order to obtain steady-state solutions to the accretion flow. The primary relations we need are the temperature gradient and velocity gradient with respect to distance from the centre, which can be integrated to obtain the full solutions for motion.

2.5.1 Governing equations

Differentiating equation (2.6) we obtain;

$$\frac{1}{\rho} \frac{d\rho}{dx} + \frac{1}{v} \frac{dv}{dx} + \frac{2}{x} = 0 \quad (2.19)$$

Now, using the definition of $\bar{e} = e/\rho = f$ and equations (2.12), (2.19) in the first law of thermodynamics (eq. (2.4)) we obtain the temperature gradient;

$$\frac{d\Theta}{dx} = -\frac{\Theta}{N} \left[\frac{1}{v} \frac{dv}{dx} + \frac{2}{x} \right] \quad (2.20)$$

Recalling the radial form of the Euler equation (eq. (2.3)), we have;

$$v \frac{dv}{dx} + \frac{1}{\rho} \frac{dp}{dx} - \frac{\lambda^2}{x^3} + \frac{1}{2(x-1)^2} = 0$$

which we integrate to obtain equation (2.5)

Comparing just the second term between (2.3) and (2.5), it is clear that;

$$\frac{dh}{dx} = \frac{1}{\rho} \frac{dp}{dx}$$

Using the definition of enthalpy $h = f + \Theta$ and definitions from equations (2.12) and (2.14), we have;

$$\left[\frac{df}{d\Theta} + 1 \right] \frac{d\Theta}{dx} = \frac{1}{\rho} \frac{dp}{dx} \quad (2.21)$$

$$N\Gamma \frac{d\Theta}{dx} = \frac{1}{\rho} \frac{dp}{dx} \quad (2.22)$$

Now using equations (2.22), (2.20) & (2.15) and rearranging terms, we obtain the velocity gradient;

$$\frac{dv}{dx} = \frac{\frac{2a^2}{x} + \frac{\lambda^2}{x^2} - \frac{1}{2(x-1)^2}}{v - \frac{a^2}{v}} \quad (2.23)$$

2.5.2 Critical points

As gravity pulls matter from afar towards the centre of an object, it accelerates. During this acceleration, matter can reach very high velocities. Very often this results

in the matter going from a subsonic velocity (slower than local speed of sound) to supersonic velocities (faster than the local speed of sound), especially in very strong gravitational potentials like that of compact objects.

Such flows are called transonic flows, and the point at which it goes from subsonic to supersonic or vice-versa (in the case of outflowing matter) is called the sonic point or critical point (often denoted as x_c). In his paper on transonic flows - [Bondi 1952](#), Bondi demonstrates that at the critical point, the velocity gradient returns an indeterminate form;

$$\frac{dv}{dx} \rightarrow \frac{0}{0}$$

This means we can derive certain conditions at the critical points, such as;

$$v_c^2 = a_c^2 \quad (2.24)$$

$$a_c^2 = \frac{x_c}{2} \left[\frac{1}{2(x_c - 1)^2} - \frac{\lambda^2}{x_c^3} \right] \quad (2.25)$$

However, this also makes calculating the velocity gradient at the critical point less trivial. By utilising the l'Hospital rule, we can determine the limiting value of the gradient at the critical point;

$$\left(\frac{dv}{dx} \right)_{x_c} = \frac{(dN/dx)_{x_c}}{(dD/dx)_{x_c}} \quad (2.26)$$

where N and D are numerator and denominator of equation (2.23), respectively.

We ultimately obtain the velocity gradient limit at the critical points in the following form;

$$A \left(\frac{dv}{dx} \right)_{x_c}^2 + B \left(\frac{dv}{dx} \right)_{x_c} + C = 0 \quad (2.27)$$

where,

$$\begin{aligned} A &= 2 + C_p \\ B &= \frac{4a_c}{x_c} C_p \\ C &= \frac{2a_c^2}{x_c^2} (2C_p + 1) + \frac{3\lambda^2}{x_c^4} - \frac{1}{(x_c - 1)^3} \end{aligned}$$

and

$$C_p = \frac{\Theta_c}{N_c + 1} \left(\frac{d\Gamma}{d\Theta} \right)_c + \frac{1}{N_c}$$

The two roots of the equation depict an inflow and an outflow solution each. Depending on the values of canonical specific energy $\mathcal{E}$ and angular momentum λ , there can be up to 3 sonic points that form in the flow. The solution topology of these critical points is different in nature. Two of them are physical "X-type" sonic points, and one is an unphysical "O-type" sonic point (Liang and Thompson 1980; Chakrabarti 1989; Chattopadhyay and Chakrabarti 2011; Kumar et al. 2013). The sonic point that forms closer to the stellar surface is called the inner sonic point (x_{ci}) and the sonic point that forms farther away is called the outer sonic point (x_{co}).

Using the critical point conditions (2.24) and (2.25), we can write the canonical specific energy equation (2.5) completely in terms of x_c and Θ_x (temperature at x_c);

$$\mathcal{E}_c = \frac{x_c}{4} \left[\frac{1}{2(x_c - 1)^2} - \frac{\lambda^2}{x_c^3} \right] + f(\Theta_c) + \Theta_c + \frac{\lambda^2}{2x_c^2} - \frac{1}{2(x_c - 1)} \quad (2.28)$$

However, since the temperature can be entirely determined by a given radius, Θ_c can be calculated, given a particular x_c .

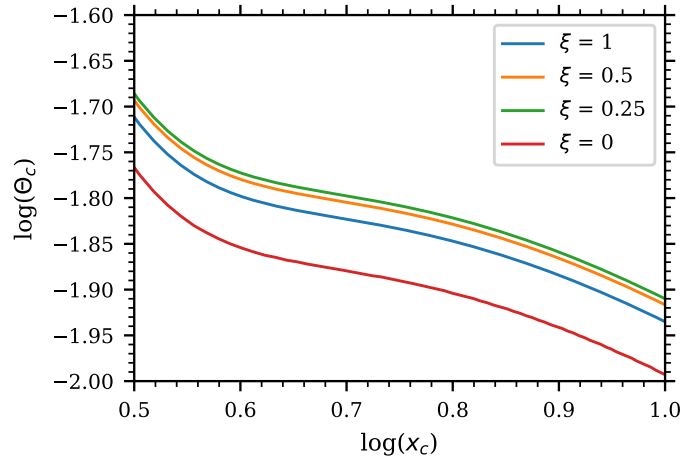


Figure 2.4: Temperature at the critical point (Θ_c) varying with critical point value (x_c) at different composition parameters (ξ)

From Fig 2.4 it is clear that as we increase the composition parameter ξ , the temperature keeps increasing but only up to $\xi \simeq 0.2$. After that, it decreases and reaches a minimum for $\xi = 0$. As the rest mass of the flow decreases with decreasing ξ , the flow becomes more energetic thermally, and this lends to the increase in temperature; however, this effect only continues till $\xi \simeq 0.2$ after which the increase in thermal energy no longer counteracts the decrease in rest mass energy (Kumar et al. 2013).

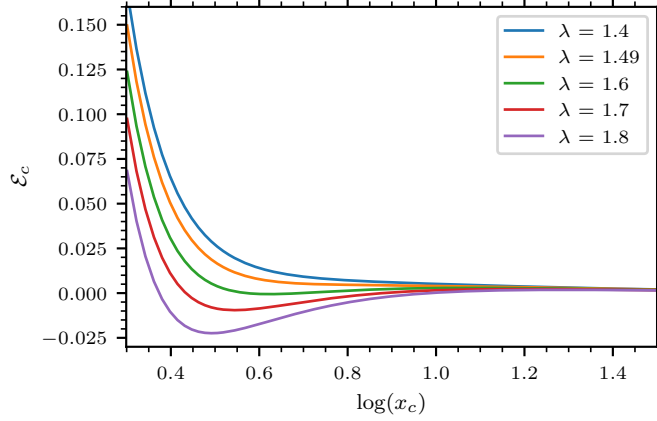


Figure 2.5: Critical energy ($\mathcal{E}_c$) varying with critical point (x_c) at different angular momentum values (λ)

Since the relation between Θ_c and x_c is known, we can determine $\mathcal{E}_c$ vs x_c . From Fig. 2.5, it is clear that there is a roughly inverse proportionality between $\mathcal{E}_c$ and x_c , i.e., the more energetic the flow is at the sonic point (by extension at other parts as well in adiabatic flows), the closer the sonic point will be to the stellar object. For a certain angular momentum, a flow can have multiple critical points. This can be ascertained from the plot by noticing that multiple x_c can have the same $\mathcal{E}_c$ values for a given angular momentum solution curve. However, below a certain angular momentum ($\lambda < 1.49$), the curve does not have a minima and, as such, does not have solutions which allow for multiple sonic points.

2.5.3 Adiabatic Shocks

In the domain of transonic accretion, shocks may originate in the flow (Chakrabarti 1989; Abramowicz and Chakrabarti 1990). The location of these shocks are dictated by the Rankine-Hugoniot conditions, which necessitate that the shock transition between two flows needs to conserve the flux of energy, momentum and mass for adiabatic shocks. Using these conditions, we can derive a relation between pre-shock (often denoted by a "-" subscript) and post-shock quantities (denoted by a "+" subscript).

The Rankine-Hugoniot condition can be represented as;

$$\begin{aligned}\mathcal{E}_+ &= \mathcal{E}_- \\ p_+ + \rho_+ v_+^2 &= p_- + \rho_- v_-^2 \\ \dot{M}_+ &= \dot{M}_-\end{aligned}$$

which, using equations (2.5) and (2.6), can be redefined as;

$$\frac{v_+^2}{2} + f(\Theta_+) + \Theta_+ = \frac{v_-^2}{2} + f(\Theta_-) + \Theta_- \quad (2.29)$$

$$p_+ + \rho_+ v_+^2 = p_- + \rho_- v_-^2 \quad (2.30)$$

$$\rho_+ v_+ = \rho_- v_- \quad (2.31)$$

Combining equations (2.29), (2.30) and (2.31), we obtain;

$$\frac{1}{v_+} [\Theta_+ + v_+^2] = \frac{1}{v_-} [\Theta_- + v_-^2] \quad (2.32)$$

$$\frac{v_+^2}{2} + f(\Theta_+) + \Theta_+ = \frac{v_-^2}{2} + f(\Theta_-) + \Theta_- \quad (2.33)$$

where $f(\Theta)$ is defined as in equation (2.9). We check where these conditions are true for obtaining the location of the adiabatic shocks in the analytical solutions.

2.6 Parameter space

The gradient functions that were obtained at the end of Section 2.5.1 can be integrated to provide a family of curves which represent either outflow or inflow solutions for a compact object. The adiabatic flows thus obtained can be constrained by a combination of 3 variables: $\mathcal{E}$, λ and ξ . Varying these parameters will yield us different kinds of flow solutions, and thus it is important to explore a wide variety of such parameters to fully understand the flow solutions for a compact object.

Connecting the extrema of the curves in Fig 2.5 and re-mapping them to an $\mathcal{E} - \lambda$ plot will give us a comprehensive overview of the whole parameter space and where each kind of solution exists in that space (Chakrabarti 1989; Kumar et al. 2013; Debnath et al. 2024).

In Fig. 2.6, the region labelled "O" yields solutions that only have an outer critical point, and the "I" region represents the solutions that only have an inner critical point. The region labelled "MCP" has multiple critical points (MCP), and solutions in this region are said to be in the MCP regime. We will be focusing on accretion solutions (inflow) for both black holes and neutron stars.

The "I" region, or the solutions with only inner critical points, exists in the space where angular momentum is relatively higher. During accretion, there is a constant

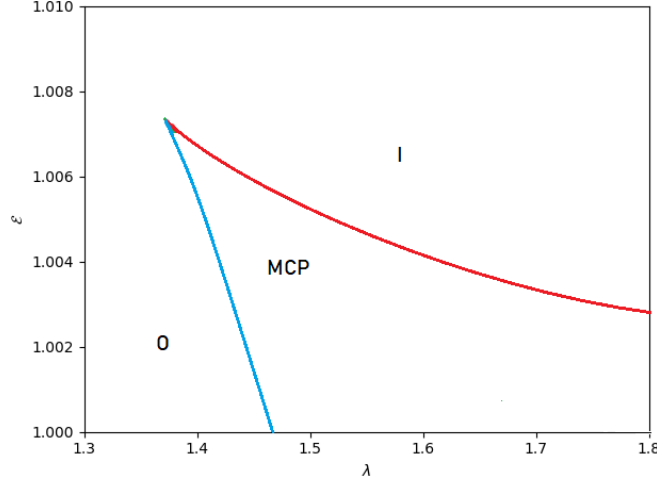


Figure 2.6: $\mathcal{E} - \lambda$ plot representing the parameter space for different solutions. The blue line is obtained by tracing the minima, and the red line is obtained by tracing the maxima of curves in Fig 2.5

tug-of-war between the centrifugal force due to rotation and the gravitational force on the flow. When angular momentum is appreciably large, the centrifugal force dominates most of the flow and prevents it from accelerating to supersonic velocity. Near the inner sonic point, the strong gravity effects of the neutron star ultimately prevail and accelerate the flow to supersonic velocity. Meanwhile, for solutions located in the "O" region with relatively lower angular momentum values, the centrifugal force is weak enough for gravity to dominate from very far away. This allows the flow to accelerate to supersonic velocities much earlier on and thus form an outer critical point.

As we will see in the coming sections, black hole accretion only presents shocks within a solution in the MCP regime, whereas neutron stars will have shocks regardless of the type of solution due to a hard surface. Nevertheless, we analyse 3 different kinds of flow solutions labelled as $P1$, $P2$ and $P3$ with varying $\mathcal{E}$ and λ while keeping $\xi = 1$

Table 2.1: Details of solution parameters for different models.

Model	ξ	$\mathcal{E}(c^2)$	$\lambda (r_g c)$
P1	1	1.0010	1.55
P2	1	1.0020	1.68
P3	1	1.0075	1.7

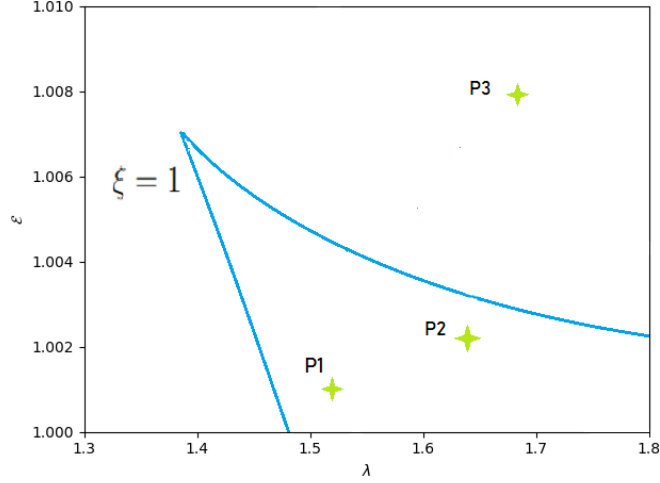


Figure 2.7: $\mathcal{E} - \lambda$ plot showing the locations of models in the parameter space.

2.7 Steady state accretion solutions

By integrating the gradient functions given by equations (2.23) and (2.20), we obtain the steady-state solutions that are required. These solutions represent both inflow and outflow conditions, but we will be focusing on the inflow solutions.

2.7.1 Methodology

It is not straightforward to integrate the gradient functions, as one needs an initial point to carry out the numerical integration. The only parameters we have complete knowledge of are $\mathcal{E}$, λ and ξ . One might as well start out with any other variable like x , v , Θ , etc., but it is quite difficult to predict what sort of solution we will obtain by constraining those variables blindly. By comparison, we can use parameter studies as outlined in section 2.6 to ascertain what kind of solution we will obtain by constraining the values of $\mathcal{E}$, λ and ξ .

Once we have fixed the three values, we need an initial value for x , as we are integrating equations (2.23) and (2.20) with respect to distance. We can obtain x from the relation outlined in equation (2.28); since at the sonic point, many independent variables become constrained due to the critical point conditions ((2.24) and (2.25)). This makes equation (2.28) entirely a function of x_c (we can obtain Θ_c from a given x_c using (2.25)). If we fix a certain $\mathcal{E}$, ξ and λ , the root of the equation provides us with the corresponding sonic point location x_c , which we can use as our initial x . Depending on whether we are solving in the MCP regime, the equation will

return multiple roots, which correspond to the multiple critical points present in the equation.

Before we start integrating the gradient functions, we also require an initial velocity v . This can be trivially obtained from the condition (2.24), where the speed of sound a can be calculated using its definition from equation (2.15) as Θ_c is already known. Since we are starting from the sonic point, we cannot use the general gradient function since it is an indeterminate form. Thus, we calculate the value of the gradient obtained from equation (2.27) by plugging in the required variables. Once we have the gradient value at the sonic point, we can start integrating from the sonic point towards the other ends of the curve, which we call a solution "branch" in this thesis.

A whole branch is integrated using the Runge-Kutta method of the fourth order (RK4), where the velocity and temperature gradients are evolved simultaneously in each time-step from an outflow radius r_{out} to the event horizon in case of a black hole and the hard surface in case of a neutron star, both denoted by r_* . In practice, the solution is integrated from the sonic point towards r_{out} and then towards r_* .

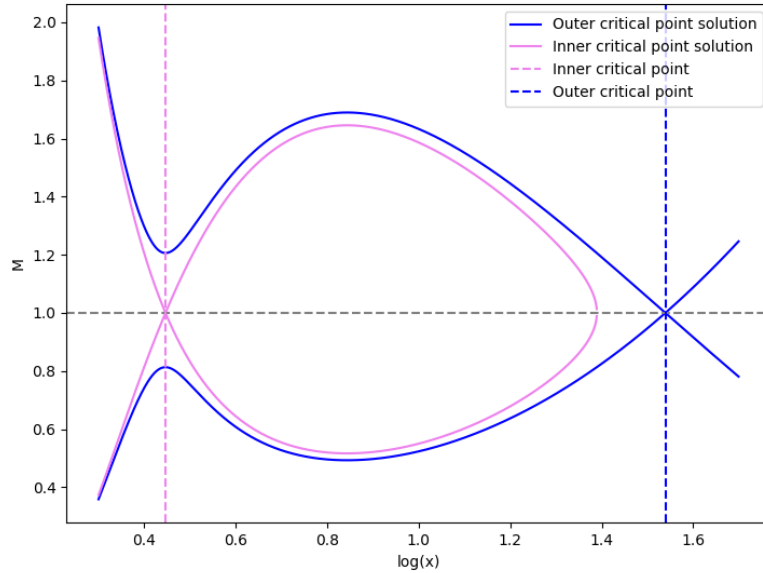


Figure 2.8: A Mach number vs logarithmic distance plot replicated from [Chakrabarti 1989](#) using parameters $\mathcal{E} = 0.689 \times 10^{-2}$, $\lambda = 1.65$. The plot shows two different branches passing through their respective critical points.

Fig 2.8 shows two branches of a solution, each branch is an isentropic curve, i.e., the entropic-accretion rate (defined in equation (2.18)) is constant throughout the curve; in principle, if we vary the entropy, we obtain a different branch. The branch which passes through the outer critical point is called the outer critical point solution, or the outer branch, and the branch which passes through the inner critical point is called the inner critical point solution, or the inner branch. Generally, the outer branch is of a higher entropy than the inner branch and after obtaining these branches with the above method, we can obtain their respective entropies using equation (2.18).

For obtaining branches that do not pass through a critical point, we need to use a different set of known quantities to start integrating the curve. Since we already have the critical point solutions, we can use their position and entropies as references. If we require a branch that is, say, at a slightly higher entropy than the outer critical branch, we can use the critical point solution as a reference and a pre-set starting distance (x) to calculate what the velocity at that initial point would be by inverting equation (2.18) for the given entropy $\dot{M}$ and solving for v . We then proceed as usual by integrating the general gradient function using the initial x and v values.

2.7.2 Black hole solutions

The solution to the equations of motion outlined in Section 2.5 is, in general, for a black hole. Thus, we will first analyse the black hole accretion solutions and then move on to exploring how it can be extended to represent neutron star accretion solutions.

Following the parameters given in Table 2.1, we obtain 3 different accretion solutions. The solutions can be roughly divided into three along the lines of angular momentum values: Low angular momentum ($P1$), medium angular momentum ($P2$) and high angular momentum ($P3$).

Fig.2.9 provides the Mach number vs logarithmic distance (in units of r_s) plots of these solutions. It provides an insight into how the accretion flow moves at various distances from the compact object. For $P1$, Fig.2.9a shows a sonic point forming at $r_c = 20.5r_s$ and for $P3$, Fig.2.9b shows a sonic point forming at $r_c = 2.5r_s$. $P2$ is an MCP solution and has two sonic points at $r_{cin} = 2.7r_s$ which is the inner critical point and $r_{cout} = 66.7r_s$ which is the outer critical point. The solution also presents two

shocks at $r = 3.7r_s$ and $r = 23.8r_s$ which can be seen in Fig.2.9c.

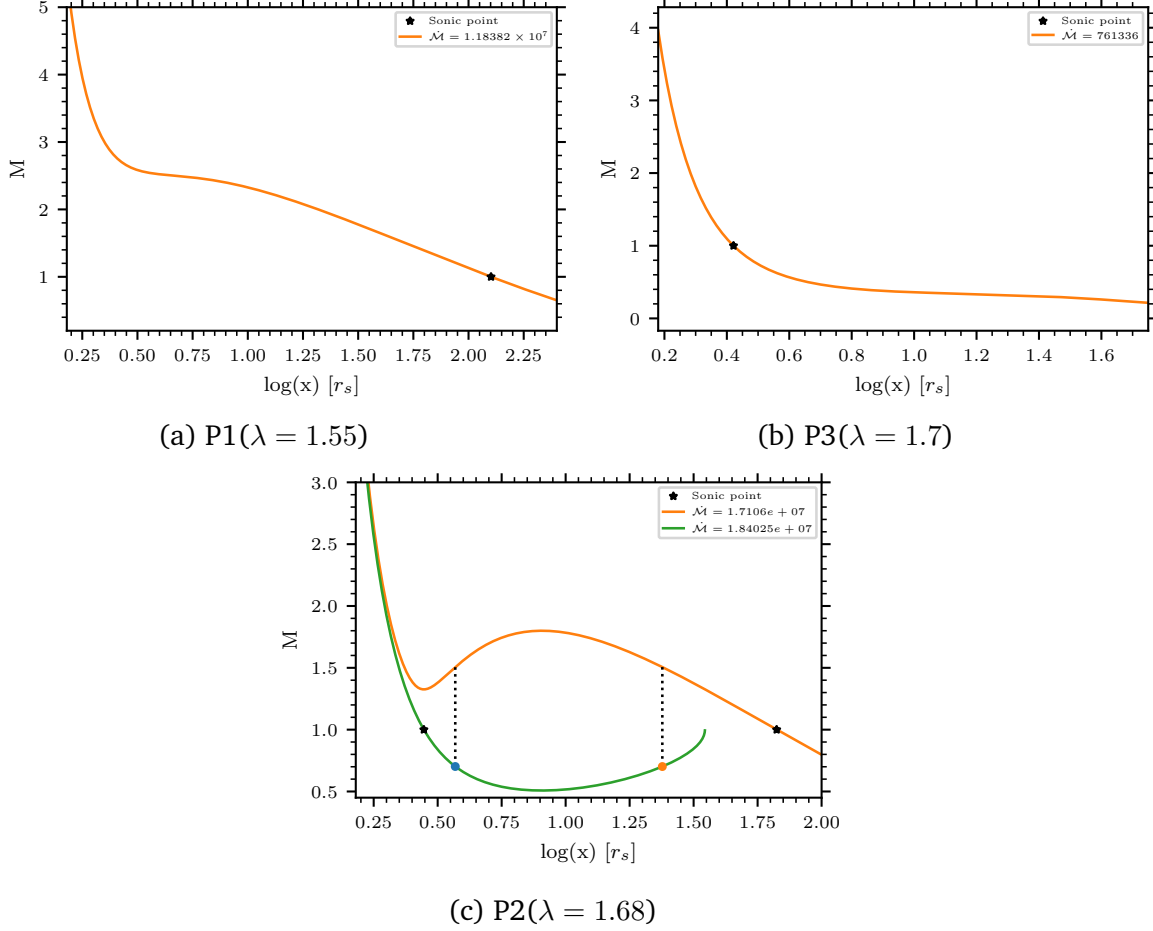


Figure 2.9: Mach number vs logarithmic distance plots of black hole solutions with $\xi = 1$ at multiple angular momentum values. Each solution corresponds to a parameter set from Table 2.1.

The solution in the MCP regime (P2) has two solution branches corresponding to the two critical points in the solution. In Fig.2.9c, the outer branch has a lower entropy than the inner branch. Black dotted vertical lines denote shock jumps between the two. These shocks are formed due to the centrifugal barrier. At a particular distance, the centrifugal force is comparable to the gravitational force and matter bunches up, forming a barrier. When gas passes through this barrier, it undergoes a shock. The flow always jumps from a lower entropy solution to a higher entropy solution through adiabatic shocks, thus it jumps from the outer branch to the inner branch.

Of the two centrifugal barrier shocks in these solutions, only the one farther away from the black hole is stable. This shock jumps onto the part of the inner branch that

is decelerating, whereas the shock that is nearer to the black hole jumps onto the part of the inner branch that is accelerating. In transonic inviscid flows, shocks that jump onto accelerating solutions are unstable (Nakayama 1992), thus the flow would in reality choose the shock farther away, which we will see in simulation results in the upcoming section. If there is a perturbation to this shock, in the case of an unstable equilibrium (accelerating part of the branch), no restoring force would act on it, and the shock would not achieve a steady state. In the case that a shock exists in stable equilibrium (decelerating part of the branch), a perturbation is countered by a restoring force, and the shock can achieve a steady state.

2.7.3 Neutron star solutions

At the beginning of the chapter, it is disclaimed that all the analytical solutions presented here are for black holes. However, it is not entirely fruitless to study neutron star accretion in this model. Accretion in neutron stars can be represented using the same global accretion solutions used for black holes, as they too are compact objects. The key difference is that since neutron stars have a hard surface, the accretion flow cannot tend to supersonic velocities near the surface. Instead, the flow must follow a solution branch that allows it to tend to subsonic velocities as it approaches the surface and eventually settle down to zero on the surface.

By extending the accretion solutions obtained for black holes in the previous section, we can analyse solutions that tend to subsonic velocities near the surface. Among the family of isentropic curves available in a particular solution, we explore a set of curves that do not pass through the critical points for this purpose by integrating at various different entropic-accretion rates. The methodology for obtaining such results is outlined in Section 2.7.1, and such a family of curves are shown in Fig. 2.10

Although these solutions are not pure neutron star solutions (since they do not contain information about the hard surface), we can study them in the context that certain shocks predicted by Rankine-Hugoniot conditions that occur near the surface allow the neutron star flows to attain subsonic velocity branches. The solutions are also terminated at a radius of $r_* = 2.4r_s$, which is expected for a neutron star of mass $M_* \approx 1.2M_\odot$, since no flow can exist beyond the surface. These shocks are created due to the supersonic flow of matter slamming onto the hard surface and developing disturbances in the flow, however, there are multiple such shocks possible as seen in Fig 2.10. The accretion flow must choose between the multiple possible

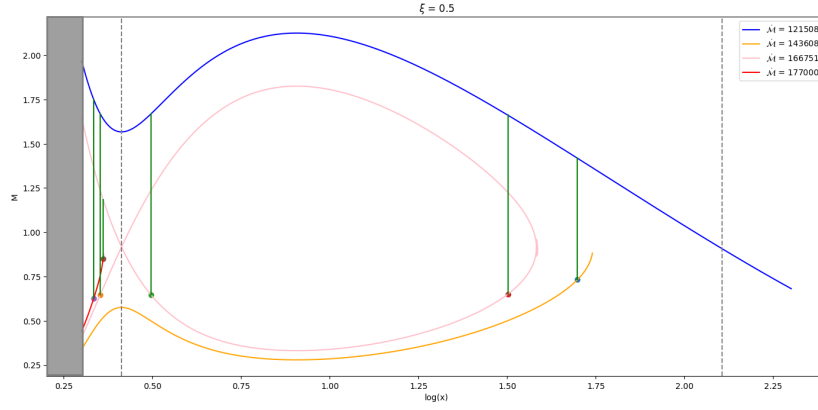
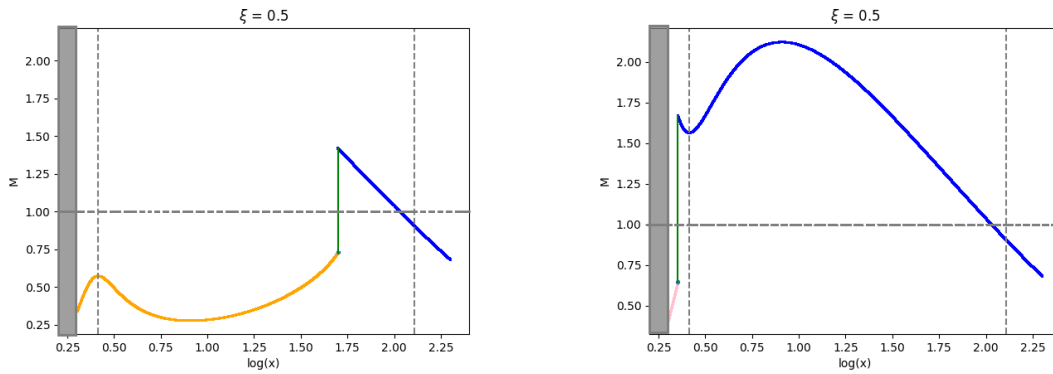


Figure 2.10: A Mach number vs logarithmic distance plot showing a family of curves in a solution using parameters $\xi = 0.5$, $\varepsilon = 1.005$, $\lambda = 1.68$. The solution also depicts all possible shock transitions predicted by Rankine-Hugoniot conditions, denoted by vertical green lines. The pink and blue curves are the typical black hole branches while the rest are associated with neutron star flows.

subsonic branches as it cannot access more than one solution at once (at least in one dimension). Notably, in these solutions, there are surface shocks which form very close to the surface and shocks which form far away from the surface, depending on which subsonic branch the flow jumps onto.



(a) Surface shock which forms far from the surface (b) Surface shock which forms closer to the surface

Figure 2.11: Plots showing different "shock routes" the flow could take depending on which surface shock is realised in the flow from fig 2.10

Fig 2.11 shows how a particular flow solution could look, depending on whether the accretion flow chooses a shock that forms far from the hard surface or closer

to the hard surface. We require hydrodynamic simulations to tell us what the flow would choose in a situation like this, which is explored in the upcoming chapter.

Fig. 2.12 presents the solutions using parameters outlined in Table 2.1. For *P1* shown in Fig. 2.12a, the outer critical branch presents its sonic point at $r_c = 20.5r_s$. This branch has an entropic-accretion rate of $\dot{\mathcal{M}} = 1.18382 \times 10^7$. A subsonic branch at $\dot{\mathcal{M}} = 2.51082 \times 10^7$ gives a surface shock at $r = 2.9r_s$ and a subsonic branch at $\dot{\mathcal{M}} = 1.84082 \times 10^7$ gives a surface shock at $r = 14.1r_s$. *P2* shows sonic point at $r_c = 66.7r_s$ in Fig. 2.12c. The two shocks for a subsonic solution at an entropic-accretion rate of $\dot{\mathcal{M}} = 1.17106 \times 10^7$ occur at $r = 2.5r_s$ and $r = 30.5r_s$. For *P3* shown in Fig. 2.12b, the shock occurs at $r = 2.45r_s$ for a subsonic solution at an entropic-accretion rate of $\dot{\mathcal{M}} = 1.89055 \times 10^7$. The sonic point for this solution occurs at $r_c = 2.5r_s$ where the critical branch has an entropic-accretion rate of $\dot{\mathcal{M}} = 1.88519 \times 10^7$.

It is also interesting to note that neutron star solutions might not show centrifugal barrier shocks. This is because the branches which allow centrifugal barrier shocks end in supersonic velocities near the surface (at $r \approx 2.3r_s$), and the surface shocks which form for such branches are only seen in Neutron stars with extremely small radii $r_* \approx 2r_s$.

For solutions with high angular momentum (*P3*), there is no "far surface shock", or at least shocks that form far enough to be termed as such. In Fig. 2.12b, we can see that the sonic point itself forms very close to the neutron star surface. Usually, for the far surface shock to form appreciably far away, the subsonic branch needs to have low enough velocities. The lower the entropy of this branch, the lower its velocity would be. However, eventually the entropy of the subsonic branch cannot go lower than the outer critical branch, which extends from infinity. As shocks only form from higher entropy to lower entropy (the second law of thermodynamics), there is a limit to how far a shock can form between these branches. As such, in the case of (*P3*), shocks beyond the sonic point can only occur for subsonic branches that have an entropy lower than the main critical branch, which would in turn show no shocks at all.

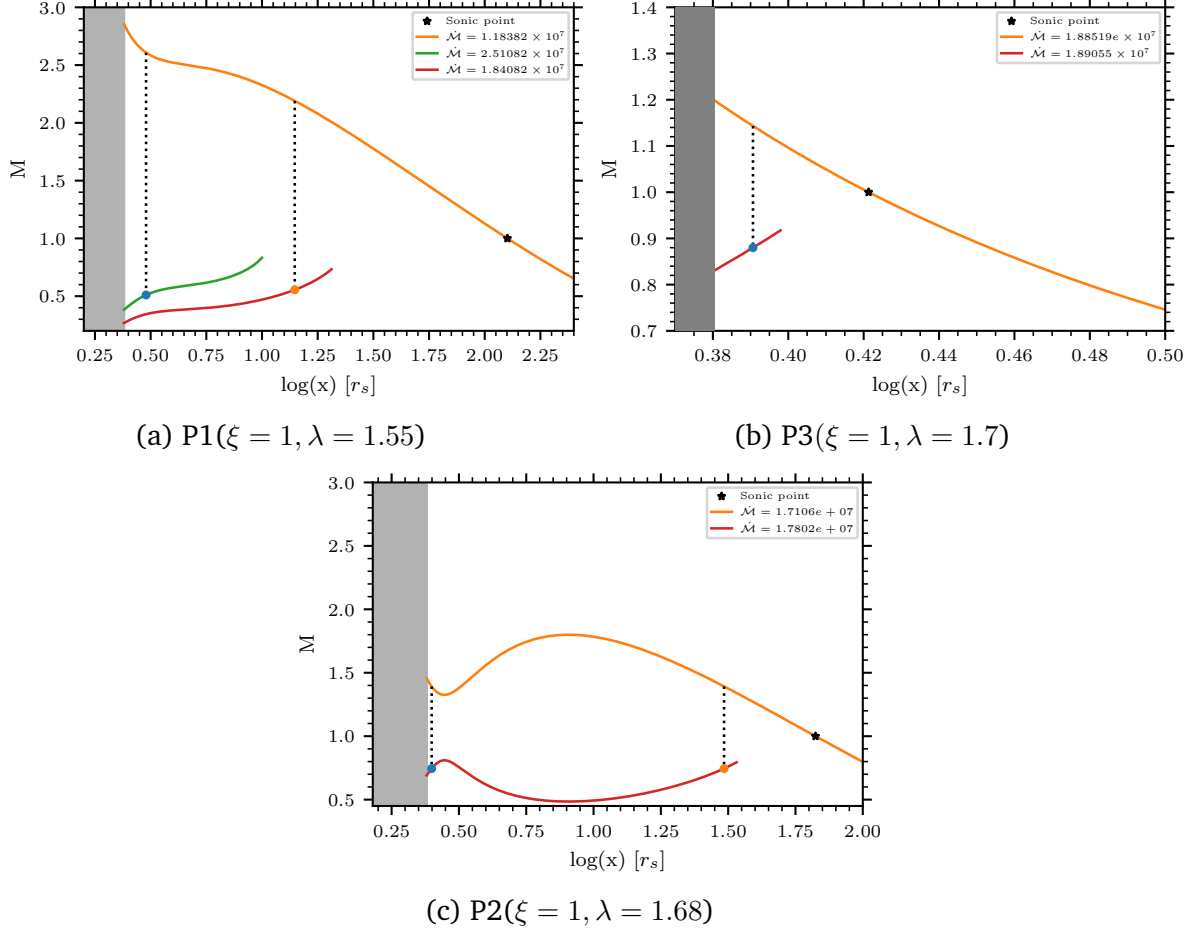


Figure 2.12: Mach number vs logarithmic distance plots of black hole solutions with $\xi = 1$ at multiple angular momentum values. Each solution corresponds to a parameter set from Table 2.1. The grey patch denotes the area inside the surface of the neutron star, which is at $r_* = 2.4r_s$

Chapter 3

Simulating accretion

The details involved in simulating accretion flows for compact objects with a hard surface are presented in this chapter. The simulation is developed in Fortran using in-house code, and the plots and simulation results are rendered in Python using standard libraries. From the multitude of solutions explored in the previous chapter, we pick two parameter sets in particular: $P2$ and $P3$. This is because $P2$ is a solution that exists in the MCP regime; this gives us the opportunity to verify the multiple shock paths predicted by the analytical solution. $P3$ is a solution with a single possible shock and acts as a way to verify non-MCP solutions.

3.1 About the simulation

The simulation that we use in this thesis is a 1-dimensional Eulerian hydrodynamic code. It solves hydrodynamic conservation equations at second order accuracy in both time and space while evolving the flow parameters of the fluid field, as opposed to a Lagrangian solver which tracks individual parcels of fluid and evolves the dynamics of fluid elements.

At its heart, the simulation is a Roe-type Riemann solver with a uniform grid size and uses the Harten scheme ([Harten 1983](#)). The Harten scheme enforces that the numerical results are positive, monotonicity preserving, and total variation diminishing (TVD). These conditions ensure that the integrated results do not create spurious maxima or minima and maintain the second-order accuracy of the calculations. Harten's TVD scheme also performs really well in the context of low dissipation and calculation speed. It uses slope limiters and flux limiters to achieve fairly non-dissipative second-order accuracy in space and then completes the second-order time update in a single step, making it computationally efficient. The upwind

finite difference that is implemented in the Harten scheme also ensures that the numerical results are physically accurate; regardless of the direction of advection, the characteristic variables are free to propagate in any direction, conserving causality. For flux reconstruction at the cell boundaries, Roe averaging is used due to its relatively low computation cost and high accuracy. The contact discontinuities at the cell boundaries and shocks are resolved by solving the Riemann problem at those locations.

This code has been used to study a range of astrophysical problems presented in works such as [Lee et al. 2011](#); [Chattopadhyay et al. 2012](#); [Raychaudhuri et al. 2021](#); [Joshi et al. 2022a](#); [Joshi et al. 2022b](#). Still, we shall use black hole solutions to validate our code and then extend it to the case of neutron star by changing the inner boundary conditions appropriately. In our simulation, we consider the following conserved quantities in the fluid flow, defined as the state variable vector $\bar{q}$ and primitive variable vector $\bar{w}$;

$$\bar{q} = \begin{pmatrix} \rho \\ M_r \\ M_\theta \\ M_\lambda \\ E \end{pmatrix} = \begin{pmatrix} \rho \\ \rho v_r \\ \rho v_\theta \\ \rho \lambda \\ \rho v^2/2 + e \end{pmatrix}; \bar{w} = \begin{pmatrix} \rho \\ v_r \\ v_\theta \\ \lambda \\ p \end{pmatrix} \quad (3.1)$$

where, ρ is the fluid rest mass density and $M_r = \rho v_r$, $M_\theta = \rho v_\theta$ are the momentum density in the r and θ directions. $M_\lambda = \rho \lambda$ is the angular momentum density, and $E = \rho v^2/2 + e$ represents the total energy density, which includes the internal energy and the kinetic energy. Here, p is the pressure; e is the energy density of the gas; v_r , v_θ , and v_ϕ are velocity components along r , θ , and ϕ while $\lambda = r v_\phi$ is the specific angular momentum. Also, $v^2 = v_r^2 + v_\theta^2 + v_\phi^2$.

The equations of motion in the conserved form for one-dimensional (1D) spherical coordinate system can be given as;

$$\frac{\partial \bar{q}}{\partial t} + \frac{1}{r^2} \frac{\partial (\bar{F}^r)}{\partial r} = \bar{S} \quad (3.2)$$

where the fluxes corresponding to the $\bar{q}$ vector are given by;

$$\bar{F}^r = \begin{pmatrix} \rho v_r \\ v_r M_r + p \\ v_r M_\theta \\ v_r M_\lambda \\ (E + p)v_r \end{pmatrix} \quad (3.3)$$

and the source term $\bar{S}$ is given by;

$$\bar{S} = \begin{pmatrix} 0 \\ \frac{\rho v_\phi^2}{r} - \frac{GM_{\text{BH}}\rho}{(r-r_g)^2} + \frac{2p}{r} \\ 0 \\ 0 \\ -\frac{GM_{\text{BH}}\rho v_r}{(r-r_g)^2} \end{pmatrix} \quad (3.4)$$

The set of equations represented by equation (3.2) consists of the continuity equation, momentum equations in three directions (r, θ, ϕ) , and the energy equation. The second row of $\bar{S}$ represents the gravitational force density, which is presented as $GM_{\text{BH}}\rho/(r - r_g)^2$, obtained from the Paczyński-Wiita potential. Moreover, $\rho v_\phi^2/r$ is the centrifugal force due to rotation. Additionally, the angular momentum equation equates to 0 since there is no viscous dissipation. The fifth row of $\bar{S}$ represents the work done due to the gravity, which is proportional to $GM_{\text{BH}}\rho v_r/(r - r_g)^2$.

Multiplying equation (3.2) by r^2 , we obtain

$$\frac{\partial(r^2\bar{q})}{\partial t} + \frac{\partial(r^2\bar{F}^r)}{\partial r} = (r^2\bar{S}) \quad (3.5)$$

$$\frac{\partial\tilde{q}}{\partial t} + \frac{\partial\tilde{F}}{\partial r} = \tilde{S}. \quad (3.6)$$

Here $\tilde{q} = r^2\bar{q}$, and accordingly the fluxes are redefined as $\tilde{F} = r^2\bar{F}^r$ and source term as $\tilde{S} = r^2\bar{S}$.

To solve the Riemann problem for this conservation equation (3.2), we can transform the conserved quantities onto the characteristic space, where the sum of the eigenvectors will give us evolution in space for the fluid flow quantities.

Generally, a homogeneous linear hyperbolic conservation equation can be written as;

$$U_t + F(U)_x = 0 \quad (3.7)$$

where U is a vector that contains the conserved fluid quantities.

The conservation equation can also be written as;

$$U_t + AU_x = 0 \quad (3.8)$$

where A is an $m \times m$ matrix (m is the number of components in U) called the characteristic matrix. For hyperbolic systems, A has m real eigenvalues, $\lambda^1, \lambda^2 \dots \lambda^m$.

Therefore, the left and right eigenvectors for this system can be written as;

$$\begin{aligned} Ar^m &= \lambda^m r^m \\ l^m A &= \lambda^m l^m \end{aligned}$$

where r^m and l^m are the m^{th} right and left eigenvectors.

Now, if we have a discontinuity in the field values, with the value on the right of the discontinuity defined as U_R and the left value defined as U_L , for the time evolution of the fluid field, we can write;

$$U(x, t) = \begin{cases} U_L & \text{for } x < \lambda^m t \\ U_R = U_L + \alpha^m r^m & \text{for } x \geq \lambda^m t \end{cases} \quad (3.9)$$

Now coming back to our simulation, the eigenvalues of equation (3.2) are given by;

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{pmatrix} = \begin{pmatrix} v_r - c_s \\ v_r \\ v_r \\ v_r \\ v_r + c_s \end{pmatrix} \quad (3.10)$$

The right eigenvectors (R) obtained are

$$R_{1,5} = \begin{pmatrix} 1 \\ a_{1,5} \\ v_\theta \\ \lambda \\ H \mp v_r c_s \end{pmatrix}; \quad R_{2,3} = \begin{pmatrix} 0 \\ 0 \\ 1, 0 \\ 0, 1 \\ v_\theta, \lambda/r^2 \end{pmatrix};$$

$$R_4 = \begin{pmatrix} 1 \\ v_r \\ v_\theta \\ \lambda \\ \frac{v^2}{2} + f - N \Theta \end{pmatrix}.$$

Here, $H = h + 0.5 v^2$. Left eigenvectors (L) corresponding to right eigenvectors are obtained from the orthogonality condition

$$R_i L_j = \delta_{ij}$$

where R_i and L_j are the components of the right and left eigenvectors, respectively. The updating of state vector $\tilde{q}^n$ to $\tilde{q}^{n+1}$ follows the procedure of [Harten 1983](#):

$$\tilde{q}_i^{n+1} = \tilde{q}_i^n - \frac{\Delta t^n}{\Delta x} \left(\tilde{f}_{x,i+1/2} - \tilde{f}_{x,i-1/2} \right), \quad (3.11)$$

$$\tilde{f}_{x,i+1/2} = \frac{1}{2} [\tilde{F}(\tilde{q}_i^n) + \tilde{F}(\tilde{q}_{i+1}^n)] - \frac{\Delta x}{2\Delta t^n} \sum_{k=1}^5 \beta_{k,i+1/2} R_{k,i+1/2}^n, \quad (3.12)$$

$$\beta_{k,i+1/2} = \mathcal{Q}_k \left(\frac{\Delta t^n}{\Delta x} a_{k,i+1/2}^n + \gamma_{k,i+1/2} \right) a_{k,i+1/2} - (g_{k,i} + g_{k,i+1}), \quad (3.13)$$

$$\alpha_{k,i+1/2} = L_{k,i+1/2}^n (\tilde{q}_{i+1}^n - \tilde{q}_{i-1}^n), \quad (3.14)$$

$$\gamma_{k,i+1/2} = \begin{cases} (g_{k,i+1} - g_{k,i}) / \alpha_{k,i+1/2}, & \text{if } \alpha_{k,i+1/2} \neq 0 \\ 0, & \text{if } \alpha_{k,i+1/2} = 0 \end{cases} \quad (3.15)$$

$$g_{k,i} = \text{sign}(\tilde{g}_{k,i+1/2}) \max\{0, \min[|\tilde{g}_{k,i+1/2}|, \text{sign}(\tilde{g}_{k,i+1/2}) \tilde{g}_{k,i+1/2}]\}, \quad (3.16)$$

$$\mathcal{Q}_k(x) = \begin{cases} x^2/4\xi_k + \xi_k, & \text{if } |x| < 2\xi_k, \\ |x|, & \text{if } |x| > 2\xi_k. \end{cases} \quad (3.17)$$

Here, $k = 1-5$ goes for the five characteristic modes. The parameters ξ_k implicitly control numerical viscosity, and they are defined for $0 \leq \xi_k \leq 0.5$. The updated fluid state variables $\bar{q}$ are obtained by incorporating the source terms with $\tilde{q}$:

$$\bar{q}_i^{n+1} = \frac{1}{r^2} \left(\tilde{q}_i^{n+1} + \Delta t^n \tilde{S}(\tilde{q}_i^n) \right) \quad (3.18)$$

To recover the primitive variables, we can use the definition of the state variables outlined in equation (3.1);

$$\begin{aligned} w_1 &= q_1 = \rho \\ w_2 &= q_2/q_1 = M_r/\rho = v_r \\ w_3 &= q_3/q_1 = M_\theta/\rho = v_\theta \\ w_4 &= q_4/q_1 = M_\lambda/\rho = \lambda \end{aligned}$$

However, recovering w_5 from q_5 is non-trivial. From equation (2.8) and equation (3.1) we can write;

$$f = \frac{E}{\rho} - \frac{v^2}{2} = C \quad (3.19)$$

where C would be known in each computational cell, using equation (2.9), we obtain a cubic equation in terms of Θ ;

$$27\Theta^3 + 9K_1\Theta^2 + 12K_2\Theta + \frac{16(1-C)}{\eta\tau^2} = 0 \quad (3.20)$$

where,

$$\begin{aligned} K_1 &= \frac{(1 + 2/\eta)(2 - \xi) + \xi(1/\eta + 2)}{\tau} + (1 - C) \\ K_2 &= \frac{2}{\eta\tau^2} + \frac{(1 - C)}{\tau}(1 + 1/\eta) \end{aligned}$$

The analytical solution to equation (3.20) can be obtained using the methods outlined in Abramowitz 1974. Defining $b_1 = 9K_1/27$; $b_2 = 12K_2/27$; and $b_3 = 16(1 - C)/(27\eta\tau^2)$, then,

$$\begin{aligned} Q &= \frac{1}{9}(3b_2 - b_1^2) \\ T &= \frac{1}{54}(9b_1b_2 - 27b_3 - 2b_1^3) \end{aligned}$$

where the discriminant will be $\Xi = Q^3 + T^2$. For CR EoS, we have $\Xi \leq 0$ and $Q \leq 0$, which tells us there are three real unequal roots. The only physical root is;

$$\Theta = 2\sqrt{-Q} \cos\left(\frac{\tilde{\theta}}{3}\right) - \frac{b_1}{3} \quad (3.21)$$

where, $\cos(\tilde{\theta}) = \frac{T}{\sqrt{-Q^3}}$

With Θ obtained, we can retrieve pressure from the definition of pressure;

$$p = \rho\Theta$$

3.2 Verifying the simulation with a black hole

To test the validity of the numerical simulation, we test it on inviscid black hole solutions. It is expected that if the simulation returns valid results, it should provide steady-state solutions that were obtained in Chapter 2, when we provide initial conditions taken from the aforementioned analytical solutions (Debnath et al. 2024).

For black hole simulations, the outer boundary conditions are set to inflow using values expected from analytical solutions of $P2$ and $P3$. We define a v_{in} and Θ_{in} for each of the models that constitute the inflow parameters starting at a distance of $r_{in} = 300r_s$

Table 3.1: Inflow values for different parameter models

Model	v_{in}	Θ_{in}
P2	1.3899×10^{-2}	4.193×10^{-2}
P3	2.1336×10^{-3}	2.9493×10^{-3}

The inner boundary is set to be an absorbing condition or sink at the event horizon. Whatever flow reaches this boundary will be completely absorbed by the boundary, as is what is expected to happen at the event horizon of a black hole. The event horizon for this black hole is set at $r_* = 1.25r_s$, and the simulation is run till a steady state is achieved.

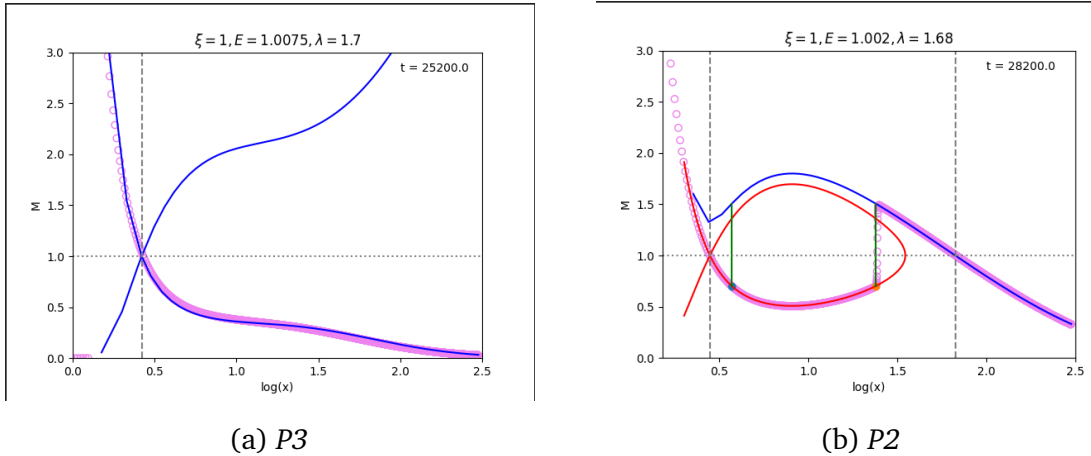


Figure 3.1: Simulation results overlaid over analytical results from $P2$ and $P3$

In Fig 3.1, the solid lines correspond to the analytical solution, while the purple

overlay corresponds to the values obtained from the simulation. It is clear that the simulation follows the analytical solution very well. For $P3$, it follows the high-angular momentum accretion solution and reaches supersonic velocities near the event horizon as seen in Fig 3.1a.

Fig. 3.1b shows us the result for an MCP solution, and it is clear that the simulation chooses the more stable one amongst the centrifugal barrier shocks as mentioned in Section 2.7.2 and then approaches supersonic velocities near the event horizon. In Fig. 3.2, one can see that as the flow comes in from r_{in} , it forms a shock near the centrifugal barrier and then stabilises to the one farther away, which is expected to be the stable shock.

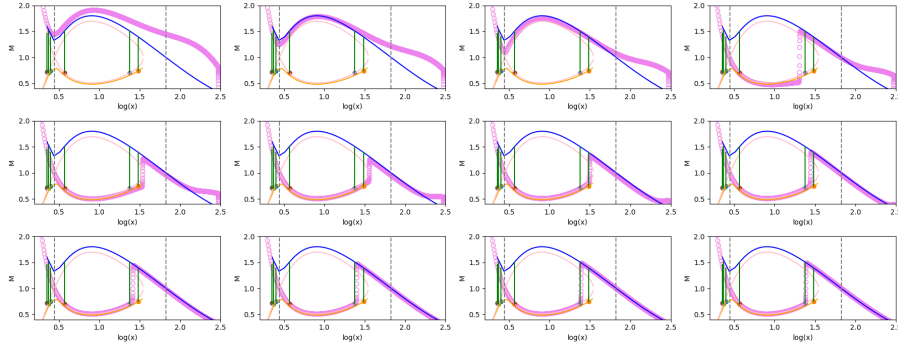


Figure 3.2: A collage of plots showing the progress of the simulation at different timesteps. Time evolves from left to right in each row, going down row by row.

3.3 Neutron star with a hard surface

Now, we simulate the accretion onto the neutron star's surface. The outer boundary condition is kept the same as was done for black hole simulations in the previous section, and the same values are used from Table 3.1. To simulate a hard surface, we implement a reflecting condition at the neutron star surface, which is set at $r_* = 2.4r_s$, which is expected for a neutron star of mass $M_* \approx 1.2M_\odot$ (Steiner et al. 2013; Suwa et al. 2018). The reflecting boundary consists of two ghost cells beyond the surface, which reflect the momentum which approaches the surface from the other side. This enforces the fact that for stable flows, the radial velocity needs to approach zero at the surface because the reflecting momenta would have effectively cancelled out.

It is to be noted that only the radial velocity component is reflected; the other

components of the velocity are copied, ensuring that unintended effects on the angular velocities do not occur. In effect, the flow interacts with a co-rotating one; there is no differential velocity in the ϕ direction at the surface.

However, on running the simulation using this setup, we do not obtain a stable solution shown in Fig.3.3. The flow becomes unstable as shocks keep generating from the surface. This can be understood by examining what is happening at the surface; as supersonic flow keeps streaming in, it is going to hit the surface and create shocks. However, since there is no dissipation that has been implemented in the model, neither radiatively nor viscously, there is nothing to stabilise the shock waves that are created. An analogy that could be made is shooting a jet of water onto a concrete floor; the water is going to keep splashing off the surface and not form a stable flow. Similarly, the supersonic flow is slamming onto the hard neutron star surface and splashing off the surface, hindering the formation of a stable flow.

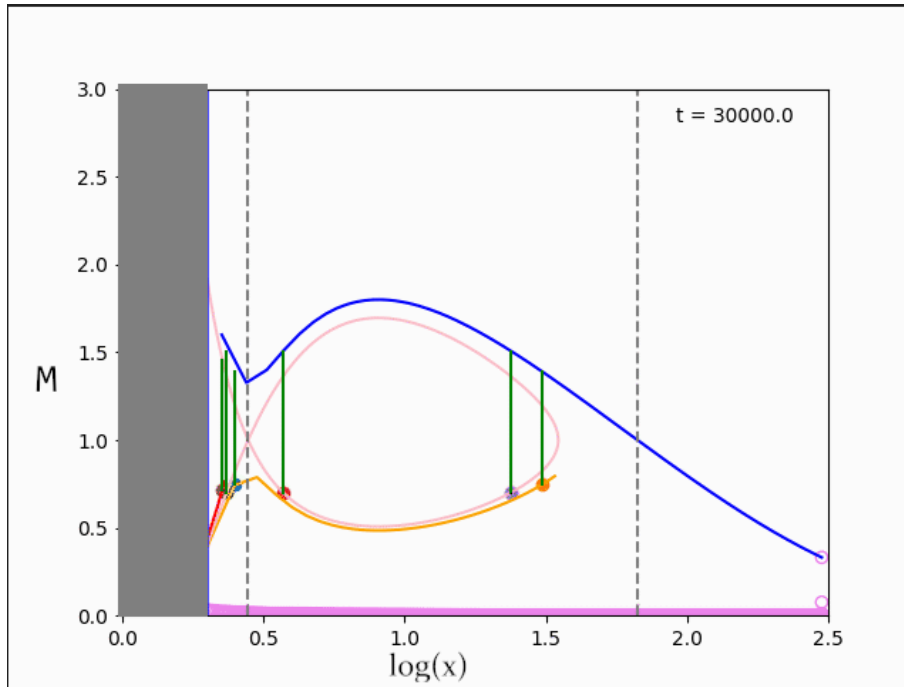


Figure 3.3: A velocity profile plot showing the absence of a stable flow solution in the simulation after implementing a hard surface. The purple simulation overlay is lying flat against the x-axis, implying a "shocked" flow.

One way we could address this is to implement energy dissipation at the surface. Since weakly magnetised Neutron stars have boundary layers, one could assume that as matter falls onto the hard surface, it starts to accumulate as it has nowhere else to go. This accumulation would lead to a massive increase in density compared to the inner parts of the accretion disk. Because of this increased density, there might be

additional dissipation at the surface that stabilises the flow. To mimic this dissipation, we implement a very simple model of cooling in the boundary layer.

Some cells above the surface are determined to contain the boundary layer (assuming the boundary layer is roughly $0.1 r_s$ tall, it would contain 3 hydrodynamic cells). Cooling is activated in only these cells; that is, some amount of velocity and temperature is reduced from these cells to mimic the effects of radiative cooling due to the boundary layer dissipation.

Defining "damping" factors $v_{\text{damp}}, t_{\text{damp}}$, we implement the cooling in this way;

$$\begin{aligned} v^{n+1} &= v^n * v_{\text{damp}} \\ t^{n+1} &= t^n * t_{\text{damp}} \end{aligned}$$

This would "damp" the velocity and temperature at each timestep (advanced by index n) by a set amount, thereby mimicking the constant dissipation of energy in the boundary layer.

Fig 3.4 shows the stable solutions that are obtained using boundary layer-only cooling. We see in Fig.3.4c that for $P3$, we get the simulation values agreeing the high angular momentum accretion solution and then exhibiting a surface shock very near to the neutron star surface. For $P2$, we obtain two different simulation results; Fig.3.4a shows the simulation following a surface shock that is far from the surface and Fig. 3.4b shows the simulation following a surface shock that is closer to the surface.

The different simulation results arise when the damping factors in the boundary layers are different. For a more effective damping factor, the energy dissipation is greater, and the accretion flow is able to have stable shocks nearer to the surface. This can be seen in Fig. 3.5 where the gas first follows the centrifugal barrier shock and then continues to go supersonic, but shortly before hitting the surface, follows a surface-induced shock.

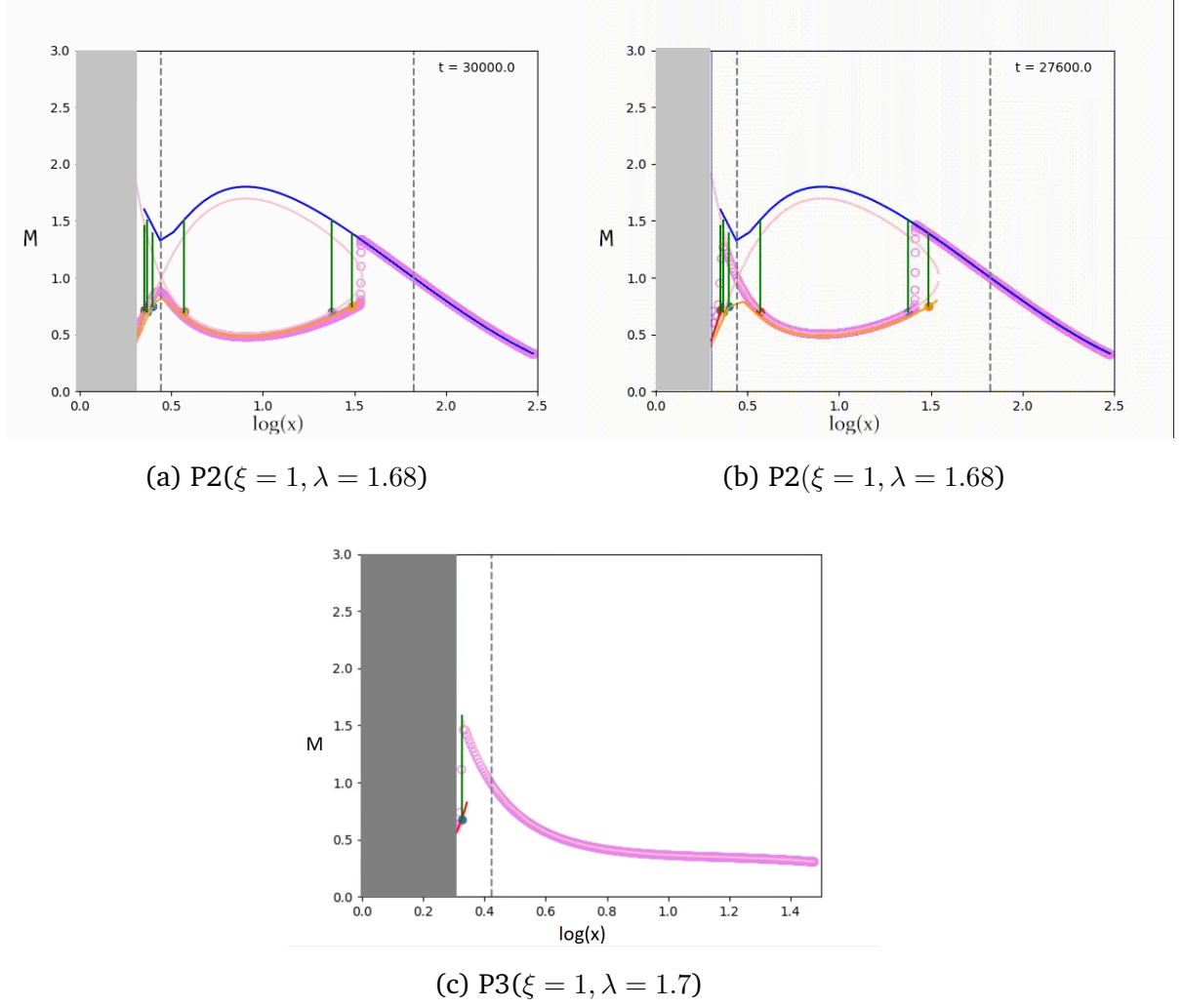


Figure 3.4: Mach number vs logarithmic distance plots of stable neutron star accretion solutions. The solid lines are analytical solutions and the purple overlay is the simulation result.

In the case of Fig. 3.6, the boundary layer damping factor is lower, leading to lower energy dissipation. This forces the accretion flow to stabilise shocks at a larger distance, and it chooses a surface shock that is far from the hard surface. Thus, depending on the extent of energy dissipation, the accretion flow chooses different solutions since the distance of stable shock formation depends on how much of the energy can be dissipated at the surface.

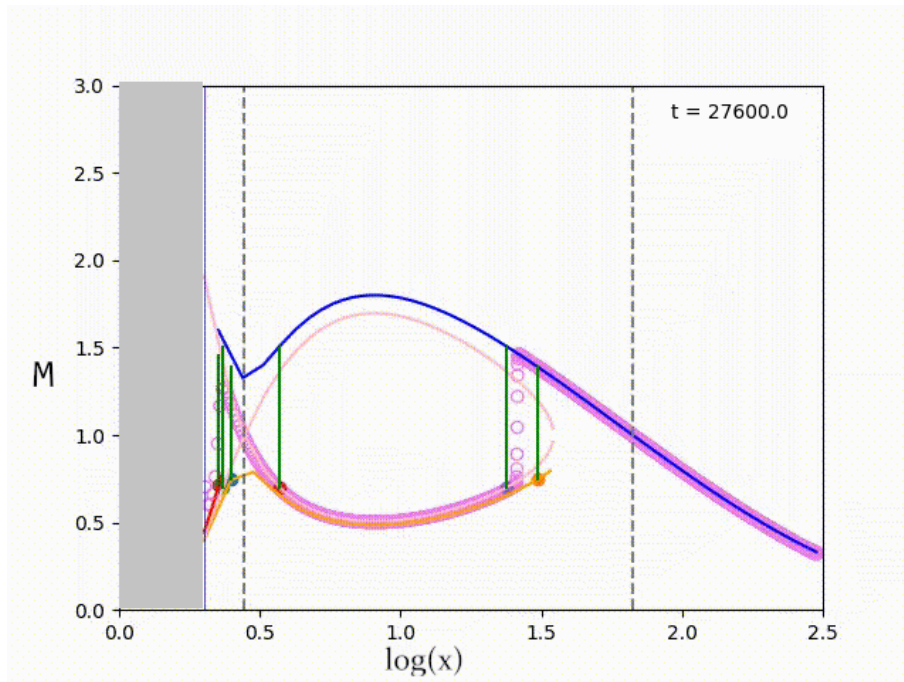


Figure 3.5: A velocity profile plot showing the simulation following the stable solution that chooses the surface shock that forms far from the surface.

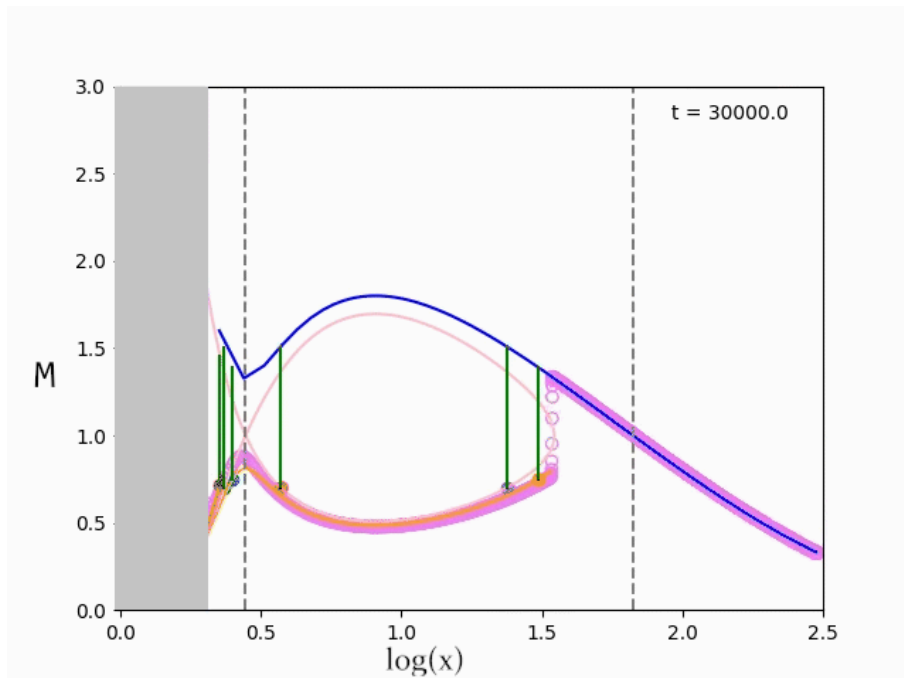


Figure 3.6: A velocity profile plot showing the simulation following the stable solution that chooses the surface shock that forms far from the surface.

Chapter 4

Conclusions & Discussion

In this thesis, we explore accretion flows onto compact objects with a hard surface using analytical and numerical methods. We first build analytical solutions corresponding to these accretion flows in Chapter 2 by using solutions for black hole accretion and then extending them to the case of weakly magnetised neutron stars with hard surfaces. We argue that the extended solution branches that are subsonic near the surface provide a valid depiction of what neutron star accretion could look like. Furthermore, we use these analytical solutions to verify our results from Numerical simulations in Chapter 3. First, we use black hole simulations to show that the simulation results are valid, and we can expect steady state solutions to be obtained as seen in our analytical calculations. We then implement a hard surface boundary condition to the simulation and use it to simulate accretion onto the Neutron star. Without implementing any energy dissipation, it is not possible to obtain any stable solutions. However, on implementing a simple cooling model in just the boundary layer of the neutron star accretion disk, we can obtain stable solutions that agree with the analytical accretion solutions. We also see that, depending on the extent of energy dissipation through the control of the "damping factor" that is used in the simulation, we obtain solutions where the accretion follows different shocks to reach the surface. More dissipation leads to accretion solutions that form shocks very close to the surface, and lesser dissipation leads to shocks that form far away from the surface.

The boundary layer is a particular region of interest because it could be the key to understanding the origin of Khz QPOs that are seen in LMXBs (Gilfanov et al. 2003; Gilfanov and Sunyaev 2014), which is often combined with a spreading layer model that attempts to explain the QPOs using instabilities on the Neutron star surface (Abolmasov et al., 2020; Abolmasov and Poutanen, 2021; Rusakov et al., 2024; Török et al., 2025). However, it is unclear whether these models can definitively explain

the X-ray variability since the models assume negligible magnetic fields even on the surface of the Neutron star. The QPOs could also be explained by oscillating shocks in the accretion, as is seen in Black hole systems ([Debnath et al. 2024](#)).

When extended to the case of weakly magnetised Neutron stars, the oscillations in the surface shocks should produce kHz frequency QPOs, given the radii at which these shocks form. A thorough spectral modelling of the X-ray emissions from the accretion flow due to bremsstrahlung and other non-thermal processes like cyclosynchrotron could give us answers. The additional modelling of other non-thermal processes is also necessitated by the fact that the flow near the surface needs to completely reach rest velocity, which is not occurring with just Bremsstrahlung cooling. This would require careful modelling of a stochastic magnetic field to implement cyclosynchrotron cooling while still keeping the analysis in the hydrodynamical regime.

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